Course Guide and Syllabus

EART23001 Atmospheric Physics and Weather

About this module

EART23001 *Atmospheric Physics and Weather* develops a physical understanding of the atmosphere, from observations and vertical structure through atmospheric dynamics, moist thermodynamics, clouds, weather systems and the global energy balance. The emphasis is on using relatively simple physical principles to interpret real atmospheric behaviour.

The weekly problem sheets are an integral part of the module rather than an optional add-on. They are designed to help you move from equations and concepts to calculations, diagrams and meteorological interpretation.

Teaching and learning

The main timetabled session is on **Thursdays, 09:00–11:00, in Schuster Moseley Theatre**. A typical two-hour session will combine:

- a focused opening lecture introducing the main physical ideas;
- time to work through selected **In Class** problems, individually or with others;
- discussion and worked examples; and
- a short wrap-up that draws the physics together and connects it to observable weather and later topics.

After the session, you should normally attempt the questions labelled **You Should Try**. Questions labelled **Extra Challenge** are optional and are there for students who would like additional depth or a more demanding application.

Teaching team and contact details

Name	Email	Location
Prof Paul Connolly	p.connolly@manchester.ac.uk	Simon Building
Dr Ian Crawford	i.crawford@manchester.ac.uk	Simon Building
Dr Hugo Ricketts	h.ricketts@manchester.ac.uk	Simon Building

Assessment and formative practice

Component	Weight	Purpose
Canvas coursework	20%	An assessed online coursework activity completed in Canvas.
Examination	80%	The main summative assessment of the module.
Example questions	0%	Unassessed practice designed to develop the methods and reasoning used in the module.
Canvas quizzes	0%	Short formative checks of your understanding, with feedback.

The example questions and Canvas quizzes do **not** contribute to the module mark. Their purpose is to give you regular practice and feedback before the assessed coursework and examination.

How to use the practice sheets

Each question has one of three labels:

- **IN CLASS**: selected anchor problems that may be worked through during the timetabled session;
- **YOU SHOULD TRY**: expected independent practice after the session;
- **EXTRA CHALLENGE**: optional additional depth for students who want it.

For numerical questions, show the physical principle or equation you are using, substitutions, units and a short interpretation of the result. For conceptual questions, aim to explain the physical chain of reasoning rather than simply state a conclusion. Worked answers are provided to help you check both the method and the interpretation.

A separate *Symbols, Constants and Units* reference is supplied. Non-standard notation is also defined locally in the questions where it is introduced.

Indicative learning outcomes

By the end of the module, you should be able to:

1. describe the basic properties of the atmosphere;
2. apply the basic laws of physics that control the behaviour of the atmosphere;
3. describe and explain the drivers for motion in the atmosphere on synoptic and global scales;
4. explain the physical behaviour of dry and moist air;
5. describe the main cloud types and cloud and rain formation processes; and
6. explain how a combination of dynamical and thermodynamical processes leads to observable weather phenomena.

Syllabus and weekly structure

Week	Date	Topic	Lecturer
1	1 Oct	Discovering the Atmosphere 1	Dr Ian Crawford
2	8 Oct	Discovering the Atmosphere 2	Dr Ian Crawford
3	15 Oct	Physics of the Atmosphere	Prof Paul Connolly
4	22 Oct	Vertical Profiles	Prof Paul Connolly
5	29 Oct	Atmospheric Dynamics	Prof Paul Connolly
6	5 Nov	Reading Week	–
7	12 Nov	Dry and Moist Air	Prof Paul Connolly
8	19 Nov	Convection and SALR	Prof Paul Connolly
9	26 Nov	Clouds and Precipitation	Dr Ian Crawford
10	3 Dec	Radiosondes	Dr Hugo Ricketts
11	10 Dec	Weather and Forecasting	Prof Paul Connolly
12	17 Dec	Global Energy	Dr Ian Crawford

A useful weekly rhythm. Attend the Thursday session, work through the selected In Class problems, complete the You Should Try questions afterwards, then use the Canvas quiz as a short check of your understanding. Return to the worked answers when you have made a serious attempt yourself.